Knuth equivalent matchings have the same core

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Editorial note. The counted statement is the author's unlabelled lemma environment at publisher-source.tex lines 1128-1130, the lemma that opens Step 1 of the proof of Theorem thm:knuth like equivalent iff same core; its proof environment opens at line 1131, so the pairing is by adjacency and the retained proof range 1131-1133 is the whole environment.

Editorial note. Because the lemma carries no author label, its identity is recorded by the pinned source position of the two environments (statement lines 1128-1130, proof lines 1131-1133) and by the claim key, which encodes that position.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the original printed number of this unlabelled lemma is not recoverable from the pinned material and is therefore not asserted.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the first step of the Knuth-equivalence criterion, stating that Knuth equivalent matchings share their core (unlabelled lemma at publisher-source.tex lines 1128-1130).

Allowed standard prerequisites (editorial).

- Knuth equivalence of matchings. Two matchings are Knuth equivalent when one can be obtained from the other by a sequence of elementary Knuth transformations.. It is the hypothesis of the statement.
- Core of a matching. The core is the matching induced on the vertices that are not removed by any reduction process.. It is the invariant the lemma claims is preserved.

Lemma 0.1. Let $m_1, m_2 \in \mathcal{M}_N$ be two Knuth equivalent matchings. Then $\mathrm{core}(m_1) = \mathrm{core}(m_2)$.

Proof. Since m_1 and m_2 are equivalent, we deduce that m_2 can be obtained from m_1 by a sequence of elementary Knuth transformations. Notably, given a matching $m \in \mathcal{M}_N$ and a matching $\varphi(m)$ obtained from m by applying an elementary Knuth transformation, the matchings m and $\varphi(m)$ differ only in the relative position of a certain short chord, and therefore $\mathrm{core}(\varphi(m)) = \mathrm{core}(m)$. A direct induction shows that $\mathrm{core}(m_1) = \mathrm{core}(m_2)$. □